

IPW3 : NACA0012 Code-to-Code Comparisons



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IPW3 – NACA0012 Case Participants

ID	Participant / Organization	Line Color
001	CIRA	
002	AeroTex	
003	NRC	
004	ONERA	
006	Dassault	
007	Polytechnique Montréal	
008	Collins Aerospace	
009	Sikorsky	
010	NASA	
013	Boeing	
014	Airbus	
015	NIAR	
019	Synopsys	
020	Bombardier	



Geometry and Aerodynamic Conditions



NACA0012 – Wind-Tunnel View

Geometry

Parameter	Value
Airfoil	NACA0012
Chord, c [m]	0.5334
Angle of attack, α [°]	4.0

Aerodynamic Conditions

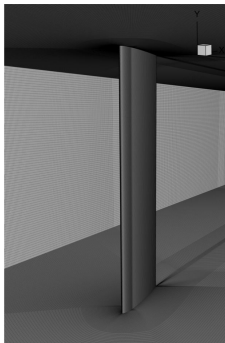
Parameter	Value
M_∞ [–]	0.3114
U_∞ [m/s]	102.15
T_∞ [K]	267.75
ρ_∞ [kg/m ³]	1.1884
Re_c [–]	3.8441×10^6

Quantities of Interest

C_L , C_D , C_M , C_p



Icing Conditions



NACA0012 – Mesh Tunnel View

Grid	Nodes	Elements
L1	60.45 M	59.86 M
L2	27.31 M	26.99 M
L3	11.23 M	11.08 M
L4	5.95 M	5.86 M

	AE3932	AE3933
T_0 [°C]	-5.6	-2.2
T_∞ [K]	262.305	265.8
M_∞	0.3151	0.3127
U_∞ [m/s]	102.31	102.19
LWC [g/m ³]	0.55	0.55
MVD [μ m]	20	20
Exposure time [min]	7	7
Exp. ice mass [g]	101.0	85.9

Experimental Observation

The warmer AE3933 condition produces approximately **15% less ice mass** than AE3932 (85.9 g vs. 101.0 g).

Droplet Distribution: 1-, 3-, 7-, and 15-bin distributions with preserved total LWC, evaluated across all four grid levels.

Quantities of Interest

β , HTC , T_s , FF , m_{water} , m_{ice} , m_{evap}



NACA0012 – Modeling Choices Across Submissions

Model / Method	001	002	004	007	008	009	010	014	019
Turbulence	$k-\omega$ TNT	$k-\omega$ SST	$k-\omega$ SST	SA	$k-\omega$ SST	$k-\omega$ SST	SA-neg-QCR	SA-neg	$k-\omega$ SST
Droplet formulation	Eul.	Lag.	Eul.	Eul.	Eul.	Eul.	Lag.	Lag.	Eul.
HTC method	–	BL Int.	T_{rec}	T_{rec}	–	–	$2 T_w$	–	–
Thermodynamics	Mess.	Mess.	Mess.	Mess.	ICE3D/EID	Fluid film	Mess.	Mess.	SWIM+
Geometry evolution	Lag./IB	Interp.	–	Lag.	ALE	–	Prism. extr.	Lag. disp.	Lag./remesh

Eul. = Eulerian; *Lag.* = Lagrangian; *SA* = Spalart–Allmaras; *SA-neg* = negative SA; *QCR* = quadratic constitutive relation; *BL Int.* = boundary-layer integral; T_{rec} = recovery temperature; T_w = wall temperature; *Mess.* = Messinger; *EID* = extended icing data; *SWIM* = shallow-water icing model; *IB* = immersed boundary; *ALE* = arbitrary Lagrangian–Eulerian; *Interp.* = interpolation; *disp.* = displacement; *Prism. extr.* = prismatoid extrusion; – = not reported.

Overall Trends

- **Turbulence:** $k-\omega$ formulations are most common (**55.6%**), followed by SA-based models (**44.4%**).
- **Droplets:** Predominantly Eulerian (**66.7%**), with **33.3%** Lagrangian.
- **Heat transfer:** Documented methods include T_{rec} (**50%**), BL integral (**25%**), and two- T_w (**25%**).
- **Thermodynamics:** Messinger-based models predominate (**75%** of documented submissions).
- **Geometry evolution:** Diverse approaches are used, with Lagrangian displacement methods most common.
- **Ice accretion:** Workshop comparisons focus on single-shot results; **4 participants** also submitted multilayer/multishot results, examined separately later.

NACA0012 – Summary of Submitted Data

AE3932

ID	L1				L2				L3				L4			
	1	3	7	15	1	3	7	15	1	3	7	15	1	3	7	15
001	✓				✓				✓				✓			
002		✓	✓	✓		✓	✓	✓		✓	✓	✓		✓	✓	✓
003					✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
004	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
006				✓				✓								✓
007	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
008		*				*				*				*		
009		*				*				*				*		
010	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
013				✓				✓								
014	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
015																
019	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
020				✓				✓	✓	✓	✓	✓				✓
Total	6	5	5	9	7	6	6	10	7	6	6	8	6	5	5	8

AE3933

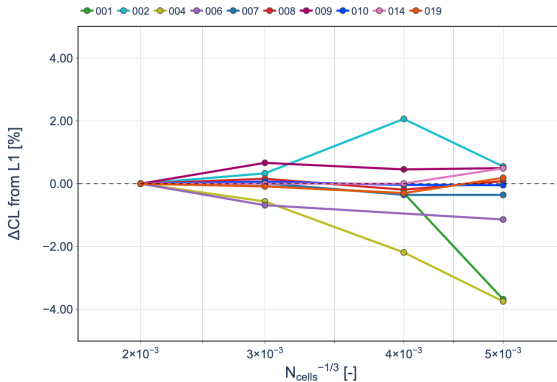
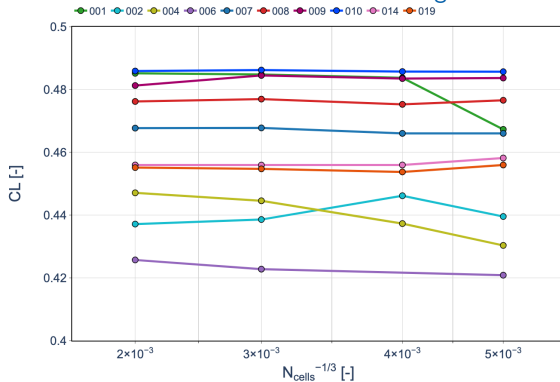
ID	L1				L2				L3				L4			
	1	3	7	15	1	3	7	15	1	3	7	15	1	3	7	15
001	✓	✓	✓	✓	✓				✓				✓			
002		✓	✓	✓		✓	✓	✓		✓	✓	✓		✓	✓	✓
003					✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
004	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
006				✓				✓								
007	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
008	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
009	✓				✓				✓				✓			
010	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
013				✓				✓								
014	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
015																
019	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
020												✓				
Total	8	6	6	8	8	6	6	8	8	6	6	7	7	5	5	6

✓: populated CutData received for the corresponding droplet-bin distribution.

*: Excel results only, no CutData slices.



NACA0012 – Lift Coefficient Grid Convergence



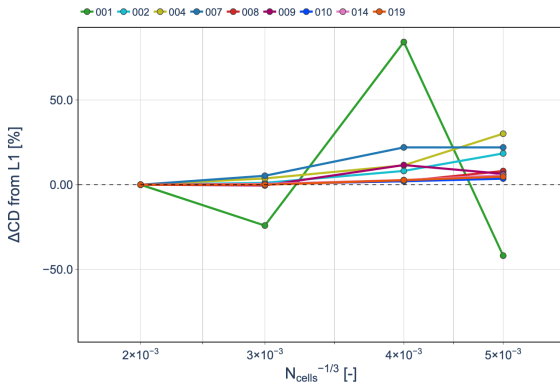
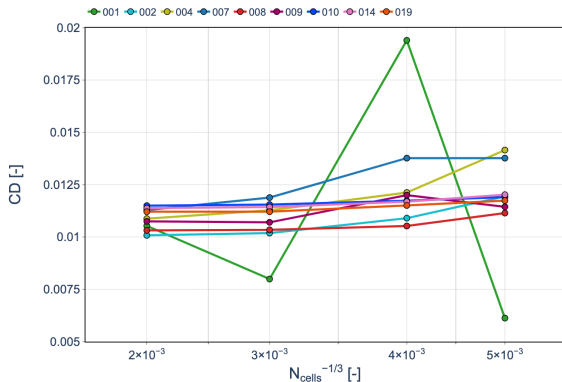
Observations

- At L1, the median lift coefficient is $C_L = 0.4677$, with an IQR of 0.0262 (5.6%).
- Most submissions exhibit weak sensitivity to spatial grid refinement.
- Relative to L1, coarse-grid differences remain within approximately 4% across the submissions.

IQR = interquartile range (75th percentile – 25th percentile).



NACA0012 – Drag Coefficient Grid Convergence

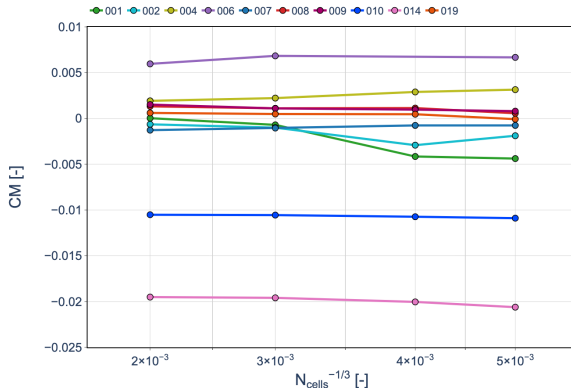


Observations

- At L1, the median drag coefficient is $C_D = 0.01088$ (**108.8 drag counts**), with an IQR of **7.57 drag counts** (7.0%).
- C_D exhibits the greatest sensitivity to spatial grid refinement, with coarse-grid deviations reaching approximately **30%** from L1.
- Most submissions show a consistent refinement trend.



NACA0012 – Pitching Moment Grid Convergence

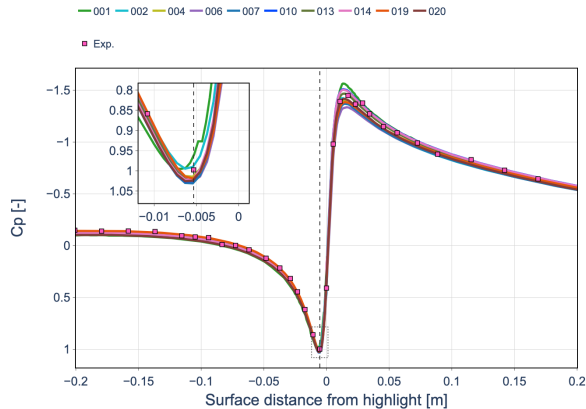
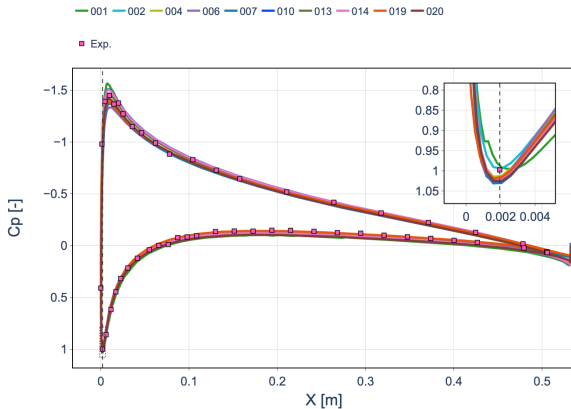


Observations

- At L1, the median pitching-moment coefficient is $C_M \approx 2.50112 \times 10^{-5}$, with an IQR of 2.60×10^{-3} .
- Most submissions exhibit little variation in C_M across the grid hierarchy.



NACA0012 – Surface Pressure Distribution (L1)



Observations

- The L1 C_p distributions are tightly clustered and show good overall agreement with the experimental measurements.
- The largest code-to-code differences are localized near the leading-edge suction peak and stagnation region.



NACA0012 – Aerodynamic Assessment Summary

Grid-Refinement Behavior

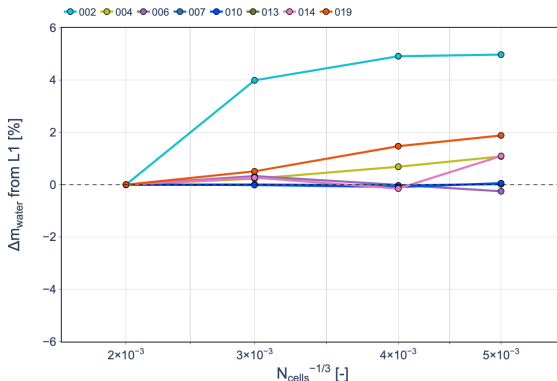
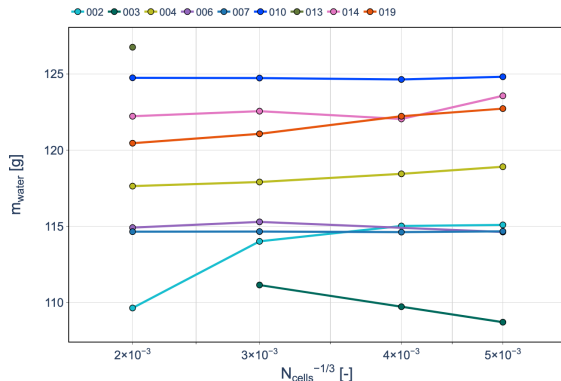
- C_L and C_M exhibit limited within-code sensitivity to grid refinement for most submissions.
- C_D shows greater and less uniform grid sensitivity, including non-monotonic behavior for selected submissions.
- For most submissions, code-to-code differences exceed the changes produced by spatial refinement.

Aerodynamic Takeaway

Overall, the aerodynamic predictions are well clustered, with L1 C_p distributions showing good agreement with experiment despite differences in prescribed surface roughness. Among the integrated coefficients, C_D exhibits the greatest sensitivity to spatial grid refinement.



NACA0012 – Water Mass Grid Convergence (15 Bins)

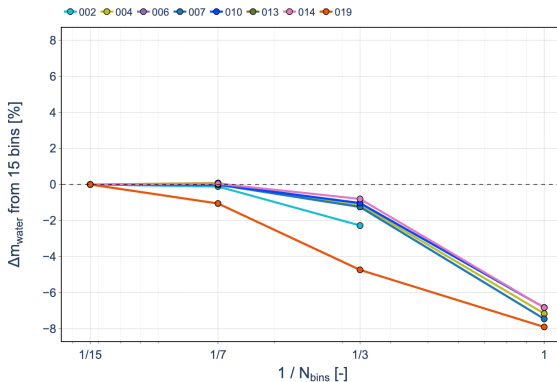
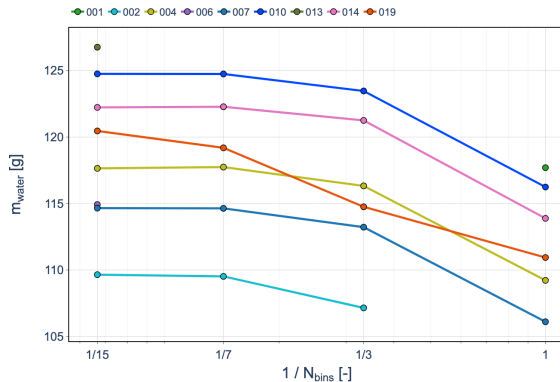


Observations

- For the 15-bin distribution, the L1 median impinged water mass is 119.05 g, with an IQR of 6.39 g (5.37%).
- Most submissions exhibit weak grid sensitivity and remain close to their L1 prediction across the grid hierarchy.



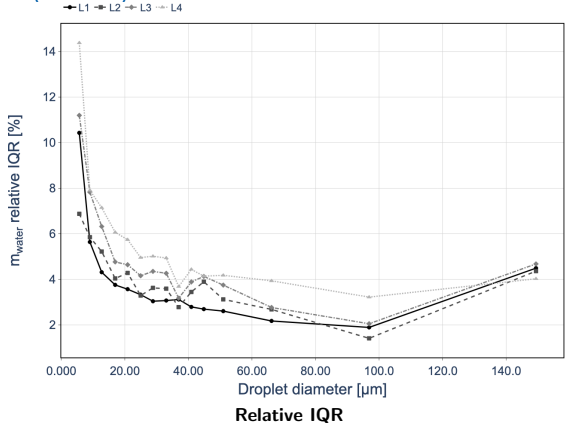
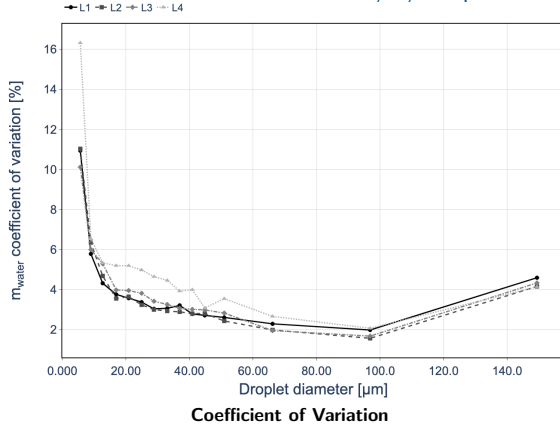
NACA0012 – Droplet Distribution Sensitivity (L1)



Observations

- The 15- and 7-bin distributions give similar impinged water masses.
- Reducing to 3 bins introduces only modest additional differences.
- A single-bin approximation reduces collected water mass by about **5–8%** relative to 15 bins.

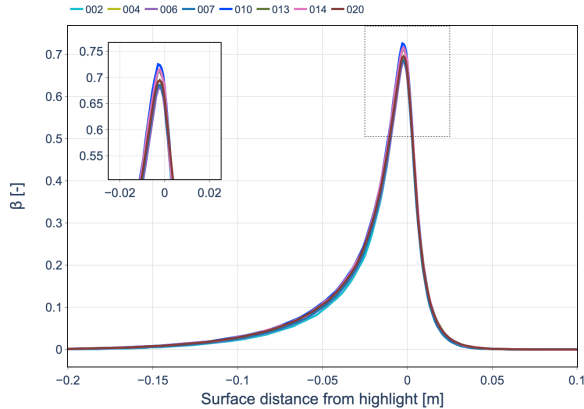
NACA0012 – Water-Mass Variability by Droplet Diameter (15 Bins)



Observations

- Variability is evaluated from the optionally submitted water mass per droplet diameter, across all available participants and grid levels.
- Code-to-code variability is greatest for the smallest droplets and lowest across intermediate diameters.
- Coarser grids generally show greater spread.

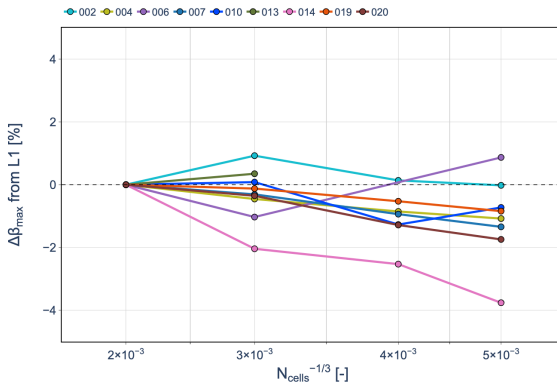
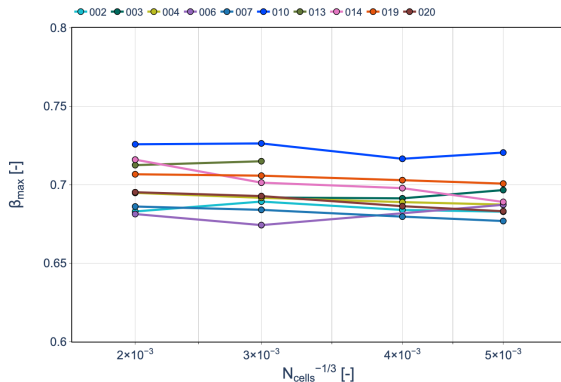
NACA0012 – Collection Efficiency (L1, 15 Bins)



Observations

- The L1 collection-efficiency profiles are tightly clustered across submissions and exhibit very similar leading-edge behavior.
- Differences are concentrated near the peak region and impingement limits.

NACA0012 – Peak Collection Efficiency Grid Convergence (15 Bins)

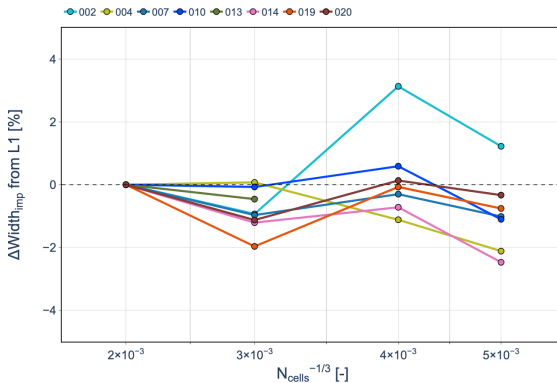
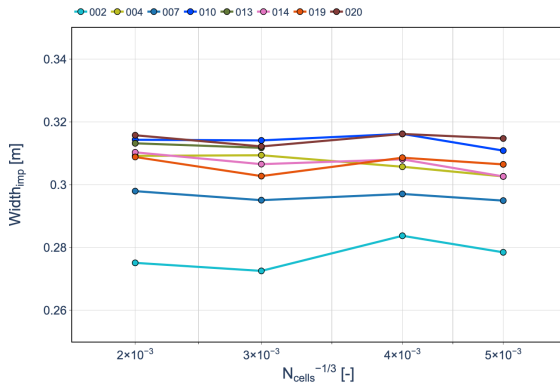


Observations

- At L1, $\beta_{\max} = 0.695$ (median), with a 3.0% IQR.
- $\beta_{\max}$ shows limited grid sensitivity, with similar trends across droplet distributions.
- Peak location is strongly influenced by leading-edge discretization; no clear grid-convergence trend is identified.



NACA0012 – Impingement Width Grid Convergence (15 Bins)



Observations

- At L1, the median impingement width is 0.309 m, with a 2.9% IQR.
- Grid sensitivity remains limited for most submissions.



NACA0012 – Droplet Impingement Assessment

Spatial Grid Convergence

- Water mass, $\beta_{\max}$, and impingement width show limited grid sensitivity for most submissions.
- Code-to-code variability is generally comparable to or larger than the grid-refinement effect.
- Peak position was examined, but its sensitivity to the local leading-edge discretization prevented a clear convergence assessment.

Droplet-Size Effects

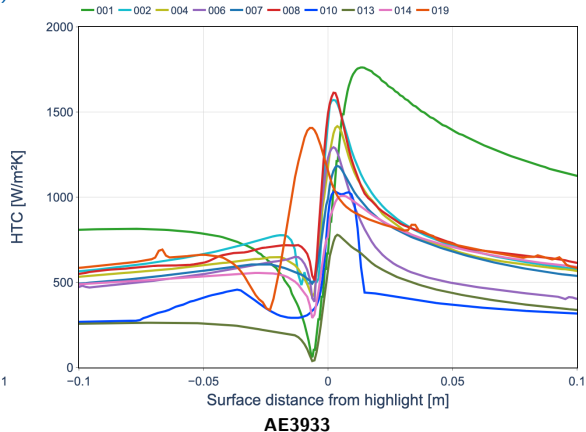
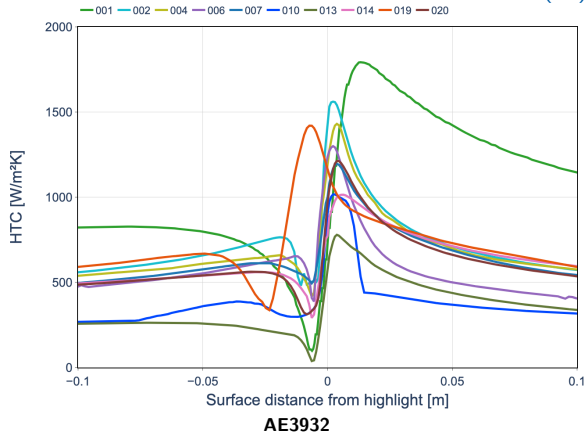
- The 7-bin distribution closely reproduces the 15-bin results, while the single-bin approximation systematically underpredicts collected water mass.
- Code-to-code variability is largest for the smallest droplets and decreases substantially over the intermediate diameter range.

Impingement Takeaway

Overall, the impingement predictions are relatively consistent with spatial grid refinement. Droplet-size discretization has a clearer systematic influence, with 7 bins providing results close to the 15-bin reference.



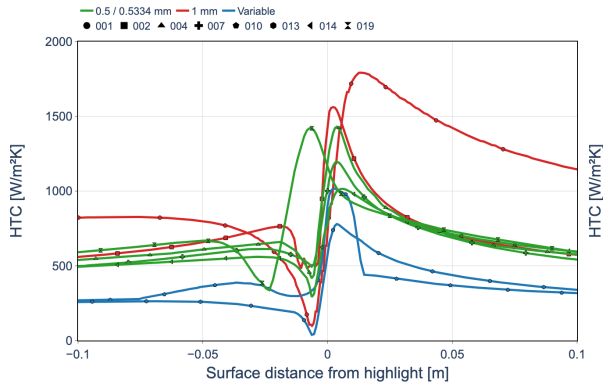
NACA0012 – Convective Heat-Transfer Coefficient (L1)



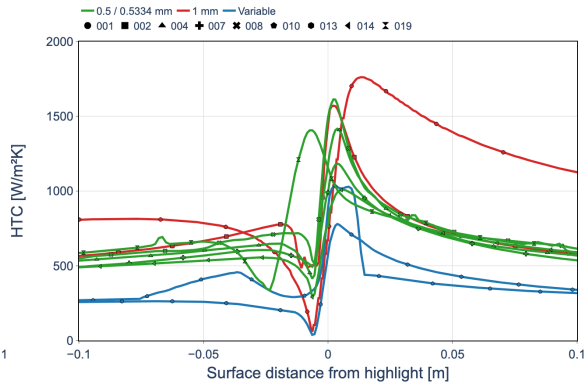
Observations

- Significant *HTC* code-to-code variability is observed, particularly near the leading edge.
- Surface roughness systematically affects *HTC*, while differences in computation methodology also contribute to the spread.

NACA0012 – Convective Heat-Transfer Coefficient (L1)



AE3932 – Grouped by Roughness



AE3933 – Grouped by Roughness

Note: Colors indicate roughness groups and are not associated with participant IDs.

Observations

- Significant *HTC* code-to-code variability is observed, particularly near the leading edge.
- Surface roughness systematically affects *HTC*, while differences in computation methodology also contribute to the spread.

NACA0012 – Integrated Convective Heat Transfer

Grid-Convergence Metric

To assess whether the differences in the local HTC distributions translate into a clear grid-convergence behavior, an integrated convective heat-transfer quantity is evaluated along the surface slice:

$$Q'_c = \int HTC(s) [T_s(s) - T_{rec}(s)] ds \quad [Q'_c] = W/m.$$

The submitted HTC and surface temperature T_s are used directly. The recovery temperature T_{rec} , however, was not part of the submitted dataset and must therefore be reconstructed.

Recovery Temperature Reconstruction

The submitted pressure coefficient is used to estimate the local external-flow state and subsequently the recovery temperature:

$$C_p(s) \longrightarrow p_e(s) \longrightarrow M_e(s) \longrightarrow T_e(s) \longrightarrow T_{rec}(s)$$

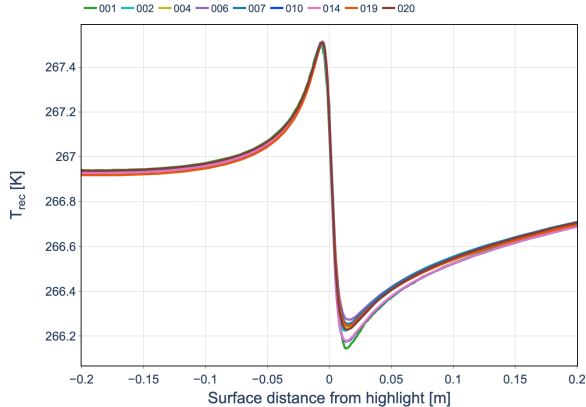
Assuming $p_s \simeq p_e$ and an isentropic external flow,

$$\frac{p_e}{p_\infty} = 1 + \frac{\gamma}{2} M_\infty^2 C_{p,e}, \quad M_e^2 = \frac{2}{\gamma - 1} \left[\left(\frac{p_0}{p_e} \right)^{(\gamma-1)/\gamma} - 1 \right],$$

$$T_e = \frac{T_0}{1 + \frac{\gamma-1}{2} M_e^2}, \quad T_{rec} = T_e \left[1 + r \frac{\gamma-1}{2} M_e^2 \right], \quad r = Pr^{1/3} \approx 0.9.$$



NACA0012 – Reconstructed Recovery Temperature (L1)

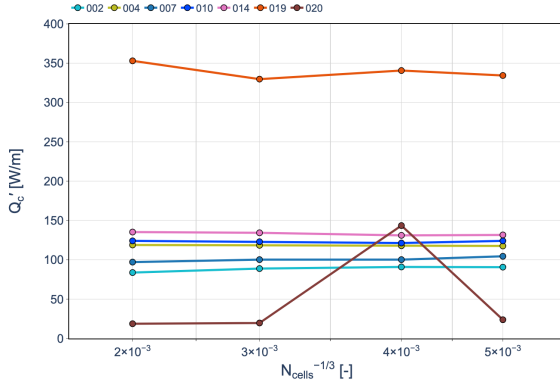


Observations

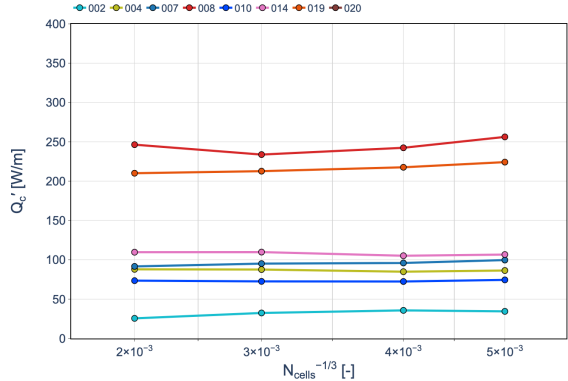
- T_{rec} is closely clustered, consistent with the agreement in C_p .
- Variability in Q'_c therefore originates primarily from HTC and T_s .



NACA0012 – Integrated Convective Heat Transfer Grid Convergence



AE3932



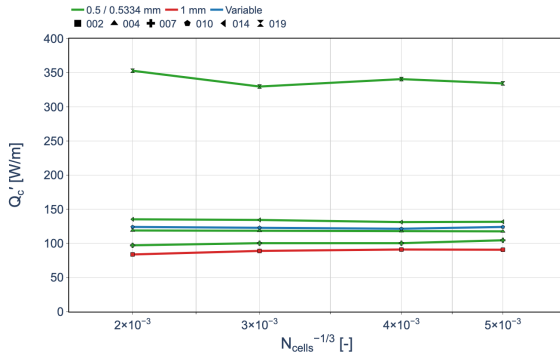
AE3933

Observations

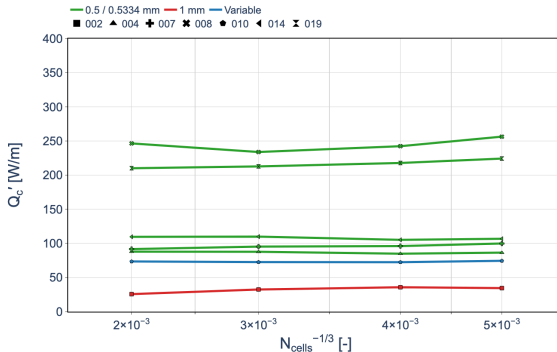
- Q'_c shows limited grid sensitivity for most submissions.
- Roughness has a code-dependent effect, with substantial variability remaining within groups.



NACA0012 – Integrated Convective Heat Transfer Grid Convergence



AE3932 – Grouped by Roughness



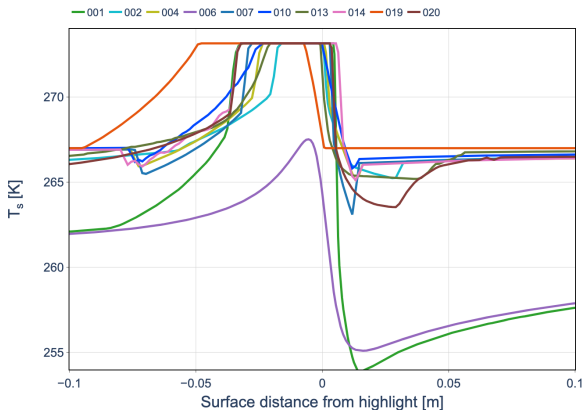
AE3933 – Grouped by Roughness

Observations

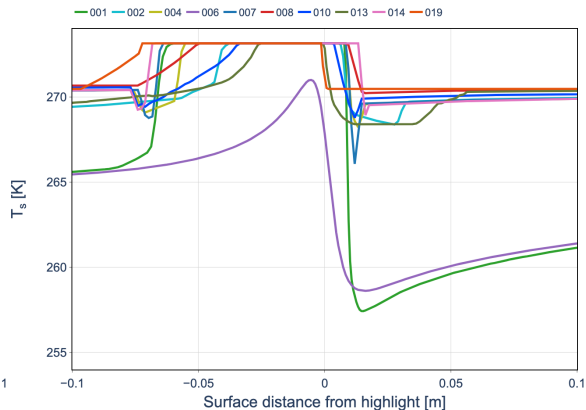
- Q'_c shows limited grid sensitivity for most submissions.
- Roughness has a code-dependent effect, with substantial variability remaining within groups.



NACA0012 – Surface Temperature Comparison (L1)



AE3932

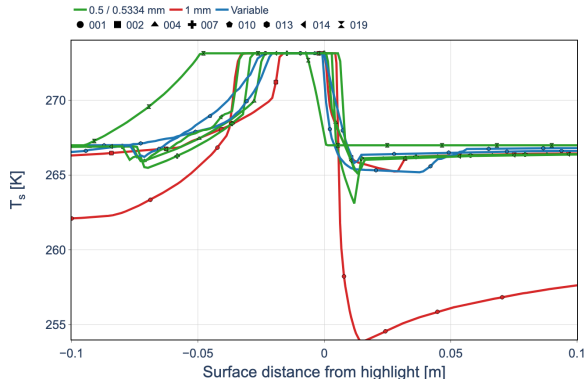


AE3933

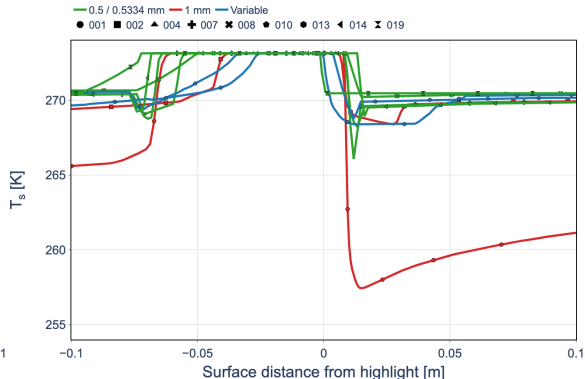
Observations

- Surface temperature exhibits substantial code-to-code variability.
- AE3933 shows a broader near-freezing region, consistent with its warmer freestream condition.
- Grouping by roughness reveals no clear systematic trend, suggesting that roughness alone does not explain the code-to-code variability.

NACA0012 – Surface Temperature Comparison (L1)



AE3932 – Grouped by Roughness

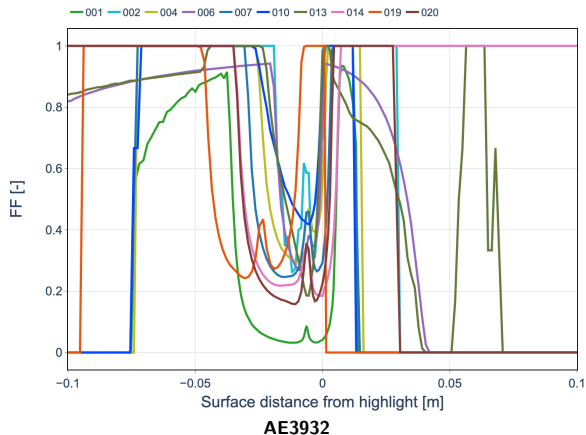


AE3933 – Grouped by Roughness

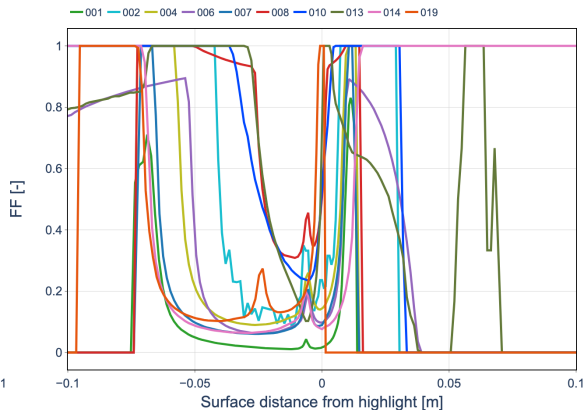
Observations

- Surface temperature exhibits substantial code-to-code variability.
- AE3933 shows a broader near-freezing region, consistent with its warmer freestream condition.
- Grouping by roughness reveals no clear systematic trend, suggesting that roughness alone does not explain the code-to-code variability.

NACA0012 – Freezing Fraction Comparison (L1)



AE3932

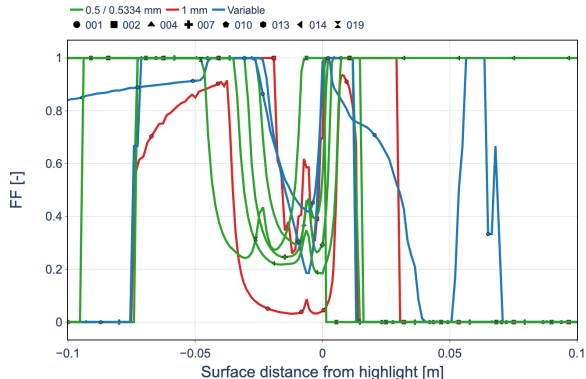


AE3933

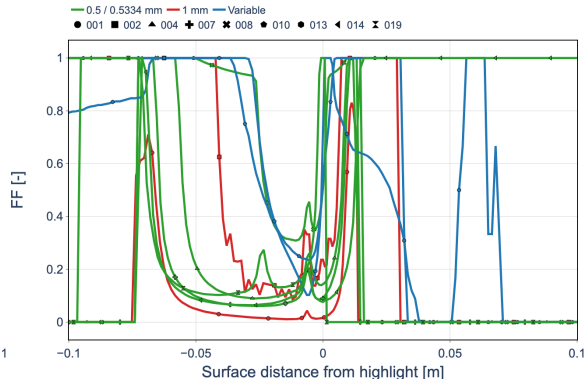
Observations

- Freezing fraction exhibits substantial code-to-code variability, with AE3933 generally showing a broader low-freezing-fraction region.
- This is consistent with the broader near-freezing surface-temperature region observed for AE3933.
- Grouping by roughness does not reveal a clear explanation for the code-to-code differences.

NACA0012 – Freezing Fraction Comparison (L1)



AE3932 – Grouped by Roughness



AE3933 – Grouped by Roughness

Observations

- Freezing fraction exhibits substantial code-to-code variability, with AE3933 generally showing a broader low-freezing-fraction region.
- This is consistent with the broader near-freezing surface-temperature region observed for AE3933.
- Grouping by roughness does not reveal a clear explanation for the code-to-code differences.

NACA0012 – HTC & Thermodynamics Assessment

Convective Heat Transfer

- Local HTC exhibits substantial code-to-code variability, with prescribed roughness explaining only a small part of the spread.
- Reconstructed T_{rec} is closely clustered, consistent with the agreement previously observed in C_p .
- Integrated Q'_c exhibits limited grid sensitivity for most submissions.

Thermodynamic Response

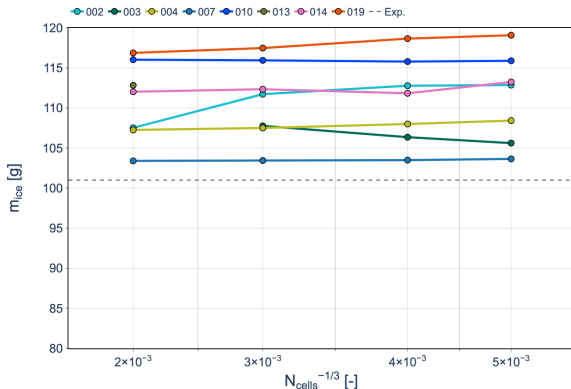
- Surface temperature and freezing fraction exhibit substantially greater code-to-code variability.
- The warmer AE3933 condition generally produces a wider near-freezing surface-temperature region and a correspondingly wider low-freezing-fraction region.

Thermodynamics Takeaway

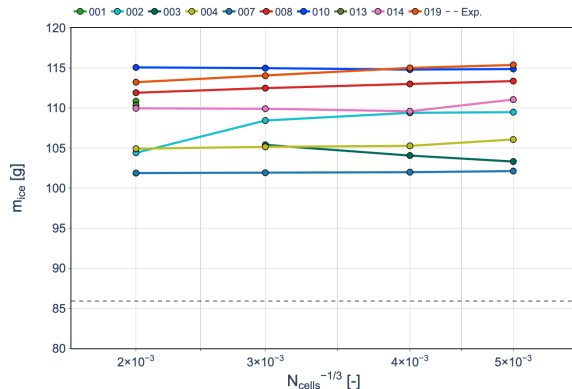
Convective heat transfer is relatively insensitive to grid refinement but exhibits substantial code-to-code variability, with roughness playing only a partial role. The warmer AE3933 condition produces a broader near-freezing region, influencing the freezing behavior and subsequent ice-accretion limits.



NACA0012 – Ice Mass Grid Convergence (15 Bins)



AE3932

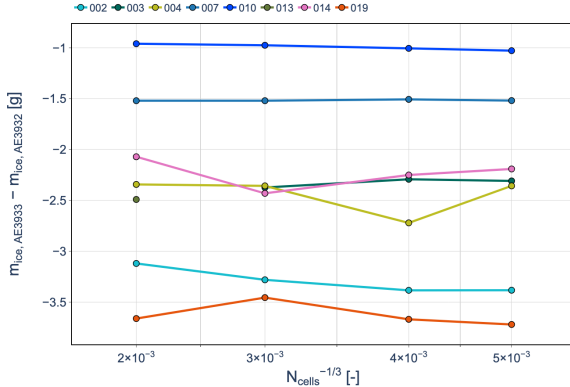


AE3933

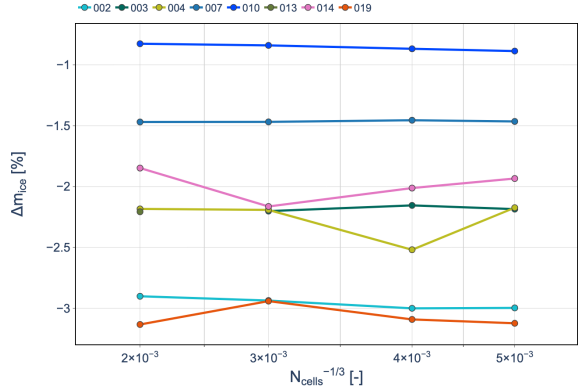
Observations

- Ice mass shows limited grid sensitivity for most submissions.
- AE3933 predicts up to $\sim 3\%$ less ice, compared with $\sim 15\%$ experimentally.

NACA0012 – Ice Mass Difference Between Conditions (15 Bins)



AE3933 – AE3932

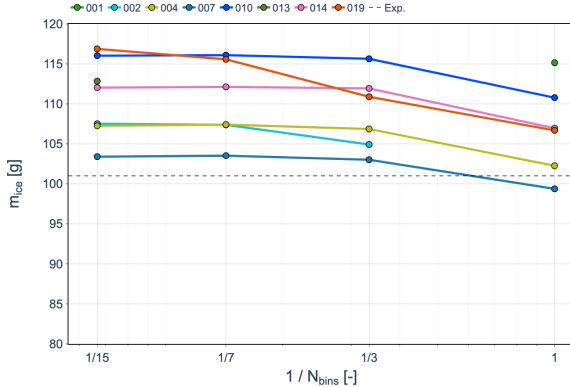


AE3933 vs. AE3932 (%)

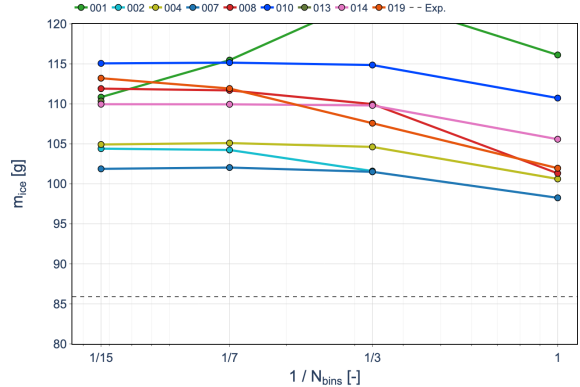
Observations

- AE3933 predicts consistently less ice for most matched submissions.
- The reduction is at most ~ 3.5 g ($\sim 3\%$), compared with $\sim 15\%$ experimentally.

NACA0012 – Ice Mass Sensitivity to Droplet Distribution (L1)



AE3932

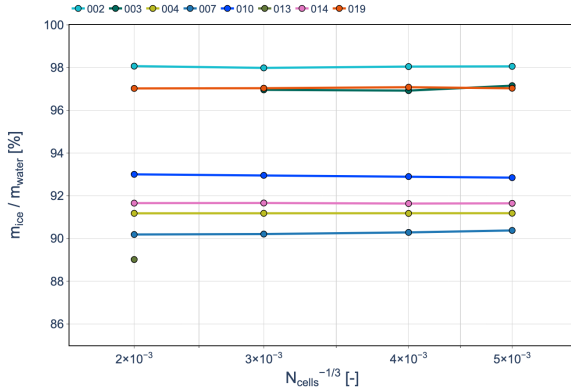


AE3933

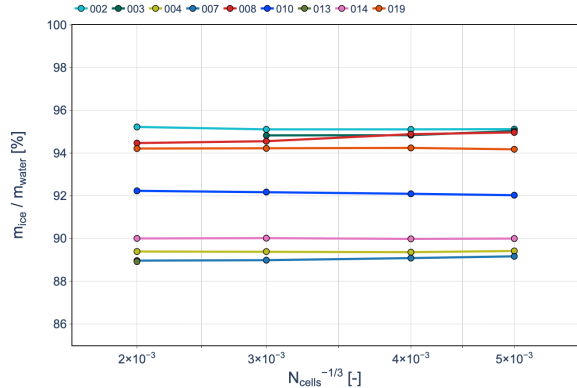
Observations

- The 15- and 7-bin distributions produce comparable ice masses.
- Sensitivity increases for 3 bins and is greatest for the single-bin approximation.

NACA0012 – Ice-to-Water Mass Ratio



AE3932

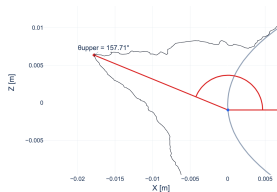


AE3933

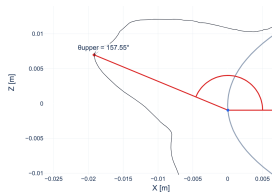
Observations

- Approximately 90% of the impinged water is converted into ice for both conditions.
- AE3933 exhibits a lower ice-to-water ratio, indicating greater water loss.
- Including evaporation closes the mass balance to approximately 100%, identifying evaporation as the primary water-loss mechanism.

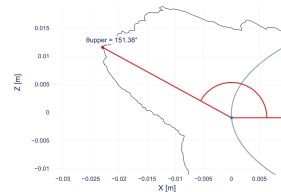
NACA0012 – Experimental Upper Horn Angle



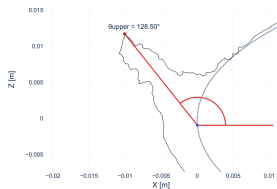
MinCCS



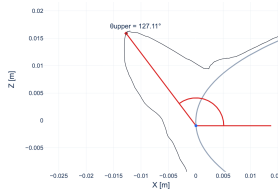
**MeanCCS
AE3932**



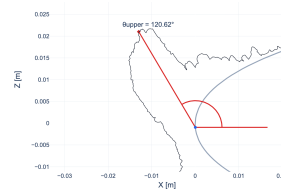
MaxCCS



MinCCS



**MeanCCS
AE3933**

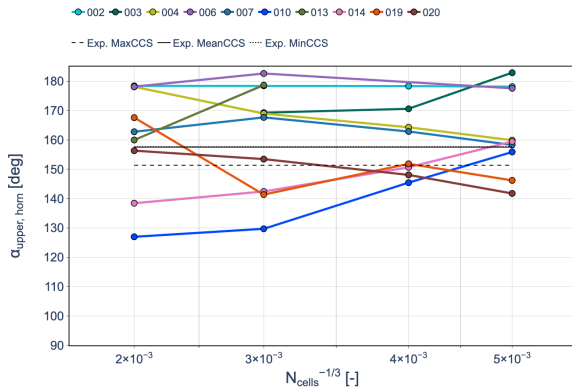


MaxCCS

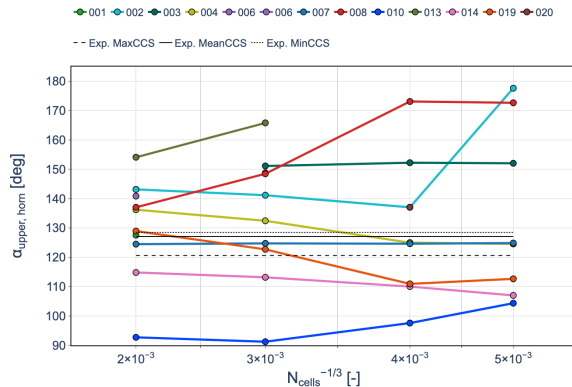
Experimental Reference

- The experimental upper-horn angle ranges from approximately $151\text{--}158^\circ$ for AE3932 and $121\text{--}129^\circ$ for AE3933.

NACA0012 – Upper Horn Angle Grid Convergence (15 Bins)



AE3932

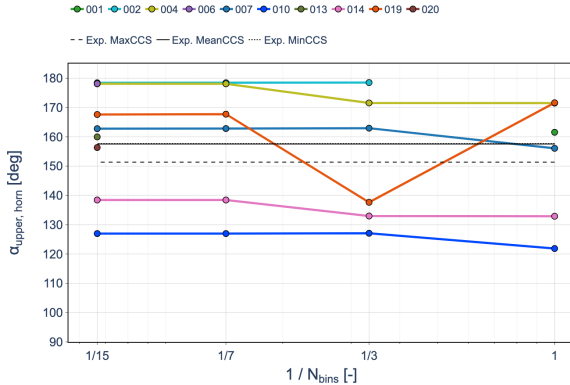


AE3933

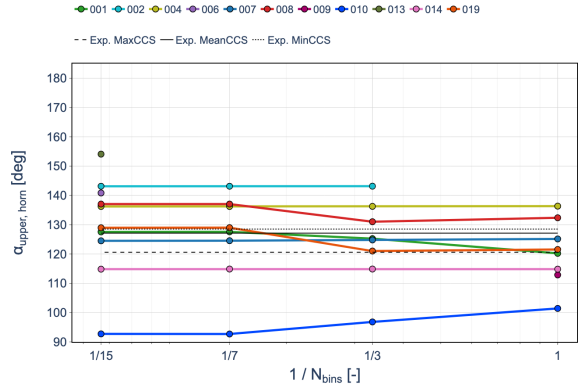
Observations

- Horn angle exhibits substantial code-to-code variability with no clear grid convergence.
- AE3933 predicts smaller upper-horn angles than AE3932, consistent with experiment.
- L1 scatter is comparable between conditions (IQR $\approx 16\%$ and 13%).

NACA0012 – Upper Horn Angle Sensitivity to Droplet Distribution (L1)



AE3932

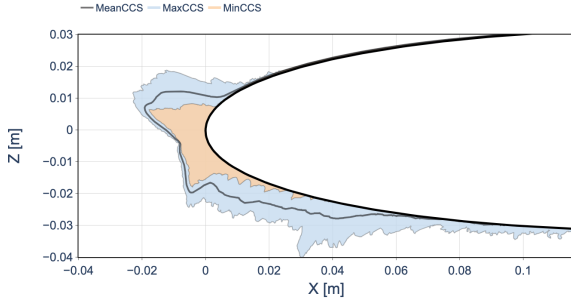


AE3933

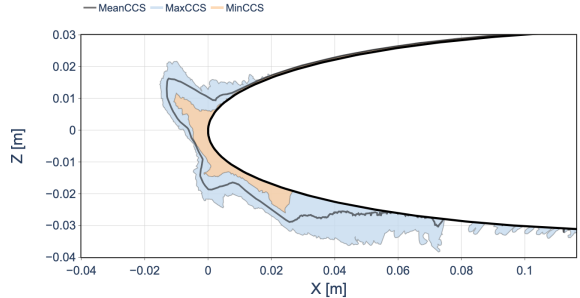
Observations

- Upper-horn angle shows limited sensitivity to droplet-size discretization.

NACA0012 – Ice Shapes (L2, 15 Bins)



AE3932 – Experiment

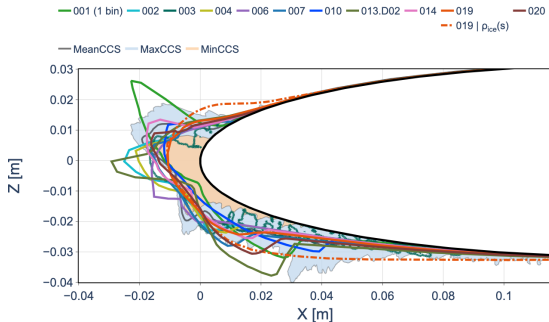


AE3933 – Experiment

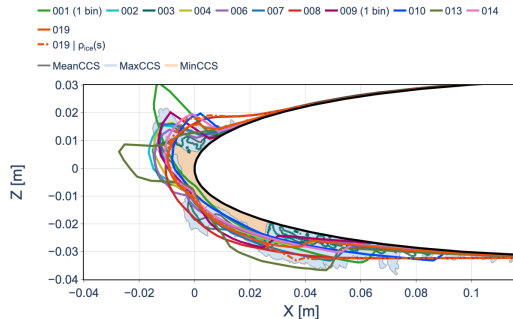
Observations

- Experiments show a clear difference in ice shape, particularly in upper-horn orientation.
- Code-to-code variability is reflected in the predicted ice shapes and horn angles.
- The experimental difference between conditions is reproduced to varying degrees.
- Roughness effects are code-dependent.

NACA0012 – Ice Shapes (L2, 15 Bins)



AE3932

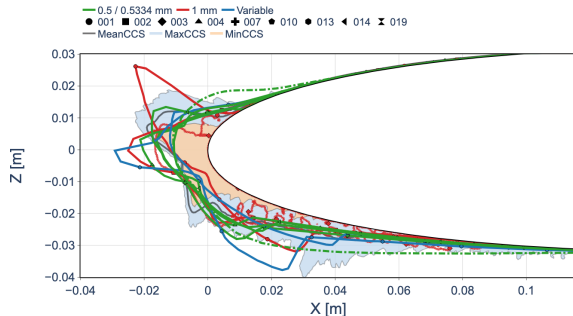


AE3933

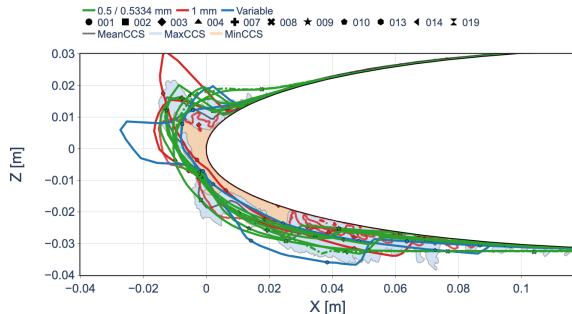
Observations

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- Code-to-code variability is reflected in the predicted ice shapes and horn angles.
- The experimental difference between conditions is reproduced to varying degrees.
- Roughness effects are code-dependent.

NACA0012 – Ice Shapes (L2, 15 Bins)



AE3932 – Grouped by Roughness

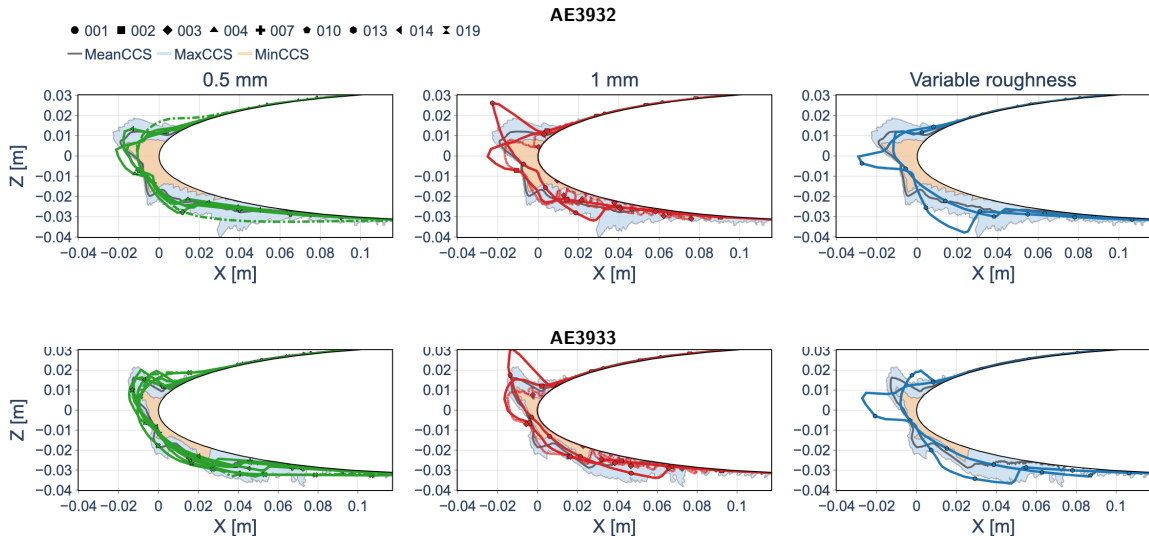


AE3933 – Grouped by Roughness

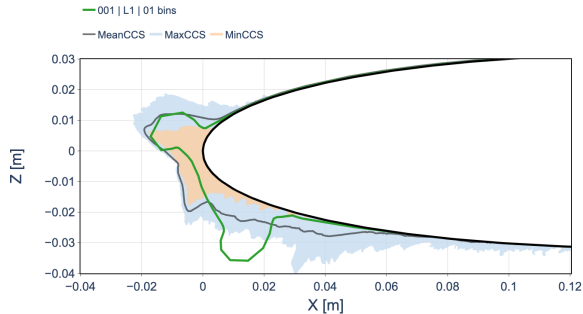
Observations

- Experiments show a clear difference in ice shape, particularly in upper-horn orientation.
- Code-to-code variability is reflected in the predicted ice shapes and horn angles.
- The experimental difference between conditions is reproduced to varying degrees.
- Roughness effects are code-dependent.

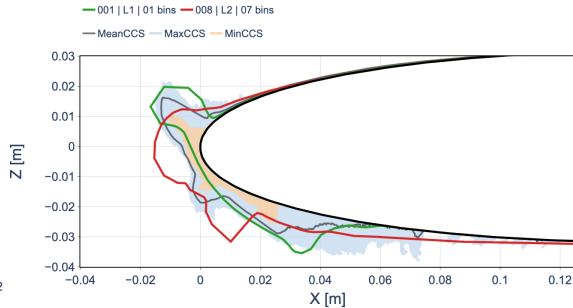
NACA0012 – Effect of Surface Roughness on Ice Shapes (L2, 15 Bins)



NACA0012 – Multilayer Ice-Shape Comparison



AE3932



AE3933

Observations

- Multilayer predictions suggest slight improvement, but the limited dataset prevents a conclusive assessment.



NACA0012 – Ice Accretion Assessment

Ice-Mass Consistency

- Ice mass shows limited grid sensitivity and good L1 agreement ($\text{IQR} \approx 5\%$); horn orientation is more variable, while 15- and 7-bin results are comparable.

Response to Icing Condition

- AE3933 predicts less ice than AE3932, but underestimates the experimental $\sim 15\%$ reduction.
- The smaller AE3933 upper-horn angle is consistent with experiment.

Ice-Shape Prediction

- Horn angle exhibits substantial code-to-code variability and sensitivity to spatial refinement, but limited sensitivity to droplet-bin discretization.
- Limited multilayer results suggest improved ice-shape agreement, but remain insufficient for firm conclusions.

Ice Accretion Takeaway

Predictions capture the experimental AE3932–AE3933 trends but underestimate the ice-mass difference, with substantial variability remaining in the final ice geometry.

NACA0012 – Overall Assessment

Overall Assessment

- **Aerodynamics:** Predictions are well clustered; C_p agrees well with experiment, while C_D shows the greatest grid sensitivity.
- **Droplet Impingement:** Limited grid sensitivity is observed, with 7-bin results closely reproducing the 15-bin reference.
- **HTC & Thermodynamics:** HTC exhibits substantial code-to-code variability, with roughness playing only a partial role; the warmer AE3933 condition produces a broader low-freezing-fraction region near stagnation.
- **Ice Accretion:** AE3933 predicts less ice and a smaller upper-horn angle, consistent with experiment, but substantially underpredicts the observed ice-mass reduction; considerable shape variability remains.

Overall Takeaway

Aerodynamics and impingement are comparatively consistent across submissions, while variability increases through thermodynamics and final ice geometry. The AE3932–AE3933 experimental trends are captured qualitatively, but their magnitude remains underpredicted.

